

Functional and differential equations

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Linear equations, systems and boundary value problems

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Theoretical minimum.

Theorem 1 (structure of the solution set). The solutions of a linear homogeneous equation of order n form an n -dimensional space, and the general solution of the inhomogeneous equation is a particular solution plus that space. The Wronskian satisfies Abel's identity $W' = -pW$ (Differentiable functions – 6), so it either vanishes identically or never vanishes.

Theorem 2 (constant coefficients and the matrix exponential). For $\dot{x} = Ax$ the solution is $x(t) = e^{At}x_0$, computed through the Jordan form; a root λ of multiplicity k contributes $t^j e^{\lambda t}$ with $j < k$, and the behaviour at infinity is governed by the real parts of the eigenvalues (Linear algebra – 3 and – 4). Variation of parameters gives $x(t) = e^{At}x_0 + \int_0^t e^{A(t-s)}b(s) ds$.

Theorem 3 (variable matrix: the Peano series). The solution of $\dot{x}(t) = A(t)x(t)$, $x(0) = x_0$, with continuous A is

$$x(t) = x_0 + \sum_{n \geq 1} \int_{0 \leq s_1 \leq \dots \leq s_n \leq t} A(s_n)A(s_{n-1}) \cdots A(s_1)x_0 ds_1 \cdots ds_n,$$

the series converging absolutely; for commuting $A(t)$ it collapses to $\exp(\int_0^t A)$.

Fact 1 (boundary value problems are different). A boundary value problem may have no solution or infinitely many: $y'' + y = 0$ with $y(0) = y(\pi) = 0$ has a whole line of solutions, while the same equation with $y(0) = 0$, $y(\pi) = 1$ has none. Uniqueness is a statement about the absence of a non-trivial solution of the homogeneous problem, that is about the eigenvalues of the corresponding Sturm–Liouville problem.

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1. Prove that the solutions of a linear homogeneous equation of order n form an n -dimensional space, prove Abel's identity for the Wronskian, and derive the formula of variation of parameters for $y'' + py' + qy = g$.
 2. (Kapacëв 4.8) Prove that the solution of the homogeneous linear system $\dot{x}(t) = A(t)x(t)$ with continuous matrix $A(t)$ and $x(0) = x_0$ is given by the Peano series written in Theorem 3, and that the series converges for every t .
 3. Prove that the boundary value problem $y'' = f(x, y)$, $y(a) = \alpha$, $y(b) = \beta$ has at most one solution whenever $\partial f / \partial y \geq 0$, and deduce the linear case $y'' = q(x)y$ with $q \geq 0$. (The vector version, $\ddot{x} = \nabla u(x)$ with strictly convex u , is problem 10 in the homework of Convex – 2.)
 4. (Kapacëв 4.9) Suppose the linear homogeneous system $\ddot{x}(t) = A(t)x(t)$ with symmetric $A(t)$ has no solution vanishing twice on $[0, \tau]$. Prove that $F(x, x) = \int_0^\tau (\langle \dot{x}, \dot{x} \rangle - \langle x, Ax \rangle) dt$ is positive for every non-zero differentiable $x: [0, \tau] \rightarrow \mathbb{R}^n$ vanishing at the endpoints.
 5. Solve the Sturm–Liouville problem $y'' + \lambda y = 0$ with $y(0) = y(\ell) = 0$, prove that the eigenfunctions are orthogonal in $L^2[0, \ell]$ and complete (Functional sequences, series and spaces – 5), and use the expansion to solve $y'' + \lambda y = g$ with the same boundary conditions.